

Remark — In Theorem 3, we assume the platform can select any network $\mathbf{P}$ it desires through its recommendation algorithm. This is without loss of generality. If we assume the social network originally begins as an arbitrary island network in Section 4, and the platform can hide and amplify content across different links, the same result readily follows.

5.2 Comparative Statics on r_P : Divisiveness and Polarization

In Theorem 3, the threshold r_P fully summarizes the extent to which misinformation will spread virally on social media. We next perform comparative statics for this threshold to understand the conditions under which the platform will create a filter bubble and propagate misinformation.

Proposition 4. *The optimal reliability threshold r_P increases as message divisiveness and/or belief polarization increases.*

Proposition 4 mimics the conclusions of Proposition 3(a). As divisiveness or polarization increases, content is consumed more aggressively within echo chambers and scrutinized more aggressively outside of them. In these settings, filter bubbles become more advantageous to the platform, especially when the relevant content has low reliability. This is because communities with more extreme beliefs now feel more strongly about news in general, rarely second-guess politically-congruent news, but often doubt and dislike counter-attitudinal news. As a result, low-reliability content would have been quickly disliked and stopped without echo chambers, which would have discouraged its sharing even by agents inclined to believe it. In contrast, inside of the platform’s filter bubbles, such content continues to spread virally.

Proposition 4 also provides a possible (albeit of course speculative) interpretation for why accelerating political polarization and identity politics in the last two decades may have come with more aggressive filter bubble algorithms from social media sites (Apprich et al. (2018)). As the recent documentary *The Social Dilemma* puts it: “The way to think about it is as 2.5 billion Truman Shows. Each person has their own reality with their own facts. Over time you have the false sense that everyone agrees with you because everyone in your news feed sounds just like you.” Tellingly in this context, while Facebook cracked down on misinformation prior to the 2020 election in part due to political pressure, its algorithms have resumed promotion of misinformation in November and December of 2020: “...the measures [Facebook] could take to limit harmful content on the platform might also limit its growth: In experiments Facebook conducted last month, posts users regarded in surveys as ‘bad for the world’ tended to have a greater reach — and algorithmic changes that reduced the visibility of those posts also reduced users’ engagement with the platform...”¹⁹

6 Regulation

Our analysis so far raises the natural question of what types of regulations might counter the viral spread of misinformation and platform choices leading to excessive ideological homophily. We

¹⁹See *Vanity Fair*:

<https://www.vanityfair.com/news/2020/12/with-the-election-over-facebook-gets-back-to-spreading-misinformation> and also <https://www.technologyreview.com/2021/03/11/1020600/facebook-responsible-ai-misinformation/>.

now briefly discuss four distinct types of regulations that have been discussed in this context: (1) regulations that force platforms to reveal articles’ *provenance*; (2) *censorship* or tagging of misinformation; (3) *performance targets* that require the platform to keep misinformation below a given threshold; (4) *network regulations*, restricting the extent of ideological homophily or segregation introduced by platform algorithms intended to maximize engagement.

We consider the effects of these policies both on a fixed sharing network and when the platform can optimally choose the sharing network given public policy. For simplicity, we suppose the regulator’s objective is to decrease the virality of articles containing misinformation on the platform.²⁰ We will say the policy has an *ambiguous* effect when there exists a sharing network (and set of conditions) where the virality of misinformation *increases* following the regulatory policy.

6.1 Provenance

We consider a policy that requires the platform to reveal the original context or *provenance* of a piece of content. For example, provenance may point the user to a peer-reviewed medical study or the full discourse from which a quote was pulled. Such a policy allows users to verify (or “fact-check”) social media content easily and quickly.

We model a provenance policy by allowing some users to fact-check the article before making their sharing and dislike decisions. In particular, we assume revealing provenance allows each agent to check the veracity of the content instantly with probability $\rho > 0$, with higher values of ρ indicating more effective provenance policies.²¹ For simplicity, we assume fact-checks perfectly reveal whether the article has veracity $\nu = \mathcal{T}$ or $\nu = \mathcal{M}$. The user can then make a more informed (but still strategic) decision about whether to share the article or take one of the other two actions.

Proposition 5. *There exists $\rho^* < 1$ such that:*

- (a) *If $\rho > \rho^*$, for any fixed sharing network, the provenance policy reduces the virality of misinformation.*
- (b) *If $\rho < \rho^*$, the provenance policy has an ambiguous effect on the virality of misinformation.*
- (c) *If $\rho > \rho^*$, the provenance policy reduces the virality of misinformation even when the platform optimally chooses the sharing network conditional on the policy.*

We see that an effective provenance policy, which helps users identify misinformation with high likelihood, successfully reduces the circulation of this content. The reason is that when it is common knowledge that a piece of content contains misinformation, incentives toward sharing are uniformly diminished across all users on the platform. Strategic complementarities then further reduce sharing. However, less effective provenance policies ($\rho < \rho^*$) are *not* always beneficial and can sometimes backfire (part (b)), as illustrated in the next example.

²⁰Formally, the regulator minimizes the virality of content conditional on the article containing misinformation; in mathematical notation, the regulator minimizes $\mathbb{E}_{\max i^*} [\mathbf{S}_{i^*} \mid \nu = \mathcal{M}]$.

²¹In practice, certain demographic groups, such as users over 65 years old, appear more likely to accept misinformation without fact-checking, perhaps because of poor media interpretation skills (see Grinberg et al. (2019) and Guess et al. (2019)). Thus, provenance policies which provide a less clear pathway to fact-checking may lead certain social media users to fact-check and others to not.